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Optical bench for spectroscopy

Abstract

Embodiments herein describe various arrangements of an optical bench used to perform spectroscopy. For example, a spectroscopy system may include a pump optical signal and a probe optical signal that are transmitted through a vapor cell on the optical bench. The optical bench can further include one or more optical components (e.g., beam splitter and a thin film polarizer) for redirecting a portion of the probe and pump optical signals to photodiodes. In one embodiment, the measurements obtained from the photodiodes can be used to perform multiple tasks. For example, the measurements can be used to adjust the power of the optical signals in the optical bench (e.g., make DC power adjustments), perform amplitude modulation correction, and lock a laser frequency to a peak of an absorption spectrum of the vapor in the vapor cell.

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Background/Summary

BACKGROUND

Field

(1) Embodiments of the present invention generally relate to an optical bench with photodiodes that measure power and modulation components of optical spectroscopy signals.

Description of the Related Art

(2) Optical atomic clocks offer improved frequency instabilities compared to microwave frequency standards due to the higher quality factor Q associated with an optical resonance. To take advantage of these high quality factors, a coherent interaction between the light and matter is required. One barrier to the widespread deployment of optical frequency standards is the complexity and cost associated with optical setups for performing spectroscopy using a vapor cell. Specifically, a typical optical bench for performing spectroscopy includes a large number of custom (and expensive) optics, polarizers, and photodiodes. Further, these optics must be stable since beam misalignment impacts the accuracy of the clock.

SUMMARY

(3) One embodiment described herein is a spectroscopy system that includes a vapor cell configured to receive a probe optical signal and a pump optical signal where optical paths of the probe and pump optical signal in the vapor cell at least partially overlap, a first photodiode configured to receive a portion of the pump optical signal before passing through the vapor cell, a second photodiode configured to at least one of (i) receive a portion of the probe optical signal after passing through the vapor cell or (ii) detect a fluorescence of the vapor cell, and a control system that is configured to perform amplitude correction and power adjustment on the pump optical signal using measurements obtained from the first photodiode and perform power adjustment on the probe optical signal and laser locking using measurements obtained from the second photodiode.

(4) Another embodiment described herein is a spectroscopy system that includes a vapor cell configured to receive a probe optical signal and a pump optical signal where optical paths of the probe and pump optical signal in the vapor cell at least partially overlap, a first photodiode configured to receive a portion of the pump optical signal before passing through the vapor cell, a second photodiode configured to at least one of (i) receive a first portion of the probe optical signal after passing through the vapor cell or (ii) detect a fluorescence of the vapor cell, a third photodiode configured to receive a second portion of the probe optical signal before passing through the vapor cell, and a control system. The control system is configured to perform amplitude correction and power adjustment on the pump optical signal using measurements obtained from the first photodiode, perform laser locking using measurements obtained from the second photodiode, and reduce noise on the probe optical signal using measurements obtained from the third photodiode.

(5) Another embodiment described herein is a method that includes measuring a first optical signal using a first photodiode before the first optical signal has passed through a vapor cell, performing amplitude modulation correction and power adjustment on the first optical signal using measurements obtained from the first photodiode, at least one of (i) measuring a second optical signal or (ii) detecting a fluorescence of the vapor cell using a second photodiode after the second optical signal has passed through the vapor cell, and performing power adjustment on the second optical signal and laser locking using measurements obtained from the second photodiode.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, may admit to other equally effective embodiments.

(2) FIG. **1** illustrates a spectroscopy system with an optical bench, according to one embodiment described herein.

(3) FIG. **2** is a flowchart for performing spectroscopy, according to one embodiment described herein.

(4) FIG. **3** illustrates optical components in an optical bench for redirecting optical signals to photodiodes, according to one embodiment described herein.

(5) FIGS. **4**A-**4**C illustrate optical components in an optical bench for redirecting optical signals to photodiodes, according to one embodiment described herein.

(6) FIGS. **5**A and **5**B illustrate integrating a vapor cell with an optical component for redirecting optical signals to photodiodes, according to one embodiment described herein.

(7) FIG. **6** illustrates a spectroscopy system with an optical bench, according to one embodiment described herein.

(8) FIG. **7** illustrates a spectroscopy system with an optical bench, according to one embodiment described herein.

(9) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

(10) Embodiments herein describe various arrangements of an optical bench used to perform spectroscopy. For example, a spectroscopy system may include a pump optical signal and a probe optical signal that are transmitted through a vapor cell (also referred to as a gas cell) on the optical bench. The optical bench can further include one or more optical components (e.g., beam splitter and a thin film polarizer) for redirecting a portion of the probe and pump optical signals to photodiodes. In one embodiment, the measurements obtained from the photodiodes can be used to perform multiple tasks. For example, the measurements can be used to adjust the power of the optical signals in the optical bench (e.g., make power adjustments), perform amplitude modulation correction, and perform laser frequency stabilization to lock the laser frequency to a peak of an absorption spectrum of the vapor in the vapor cell. Thus, while other implementations rely on three, four, or more photodiodes and complicated optical arrangements, the embodiments herein can obtain the required measurements using only two photodiodes and with a simplified optical arrangement in the optical bench (although some embodiments can use more than two photodiodes such as the embodiment in FIG. **7**).

(11) FIG. **1** illustrates a spectroscopy system **100** with an optical bench **115**, according to one embodiment described herein. In this example, the optical bench **115** receives a probe optical signal **105** and a pump optical signal **110**. In one embodiment, the probe optical signal **105** is an unmodulated optical signal while the pump optical signal **110** has already been modulated. That is, the pump optical signal **110** may have first passed through an optical modulator before being received by the optical bench **115** while the probe optical signal **105** has not. In one embodiment, the probe and pump optical signals **105**, **110** are generated by the same laser source (where the laser is split to form the two optical signals, and the pump optical signal **110** is separately modulated). Alternatively, the probe and pump optical signals **105**, **110** can be generated using separate laser

sources.
(12) In one embodiment, the probe and pump optical signals **105**, **110** are transmitted on fiber optic cables before reaching the optical bench **115** which then transmits these signals in free space. As shown, the probe optical signal **105** passes through an additional polarizer **130** which establishes fixed polarization for the optical system to reduce polarization errors in a thin film polarizer **150**. In any case, the polarizer **130** is optional and can be omitted from the system **100**.
(13) The probe optical signal **105** then travels through a vapor cell **155**. In this embodiment, the mirrors **160** redirect the probe optical signal so that it passes through the vapor cell **155** twice. However, this is just one example. In another embodiments, the probe optical signal **105** can pass through the vapor cell **155** only once, or more than two times by adding additional mirrors.
(14) After passing through a polarizer **125**, a beam splitter **145** redirects a portion of the pump optical signal **110** to a residual amplitude modulation (RAM) photodiode (PD) **165**, which will be discussed later. In one embodiment the beam splitter **145** is non-polarizing to reduce polarization sensitivity of the photodiode measurement. The remaining portion of the pump optical signal **110** passes through the beam splitter **145** and through a thin film polarizer (TFP) **150**. That is, because of the polarization of the pump optical signal **110**, it is able to pass through the TFP **150** and then enter the vapor cell **155**.
(15) The vapor cell **155** is a container (e.g., a tube) that contains atoms or molecules (e.g., Iodine or Rubidium) that have a well-defined absorption spectrum. At least a portion of the sides or walls of the vapor cell **155** is transparent so that optical signals can enter and leave the cell **155**. Light absorption or fluorescence can be measured while changing the wavelength of the pump optical signal **110** being transmitted through the vapor cell **155** which results in a peak or a series of peaks or dips in the detected signal. At a peak, the pump optical signal **110** is absorbed by the vapor in the cell **155** while at other wavelengths, the optical signal **110** passes through the vapor cell **155** with minimal absorption.
(16) As shown, the optical path of the probe optical signal **105** at least partially overlaps or is aligned with the optical path of the pump optical signal **110** in the vapor cell **155**. To perform Modulation Transfer Spectroscopy (MTS), a control system **175** detects when the modulation on the pump optical signal **110** is transferred to the unmodulated probe optical signal **105** when passing through the vapor cell **155**. When the wavelengths of the optical signals **105** and **110** are not at a peak of the absorption spectrum of the vapor in the vapor cell **155**, little to none of the modulation in the pump optical signal **110** is transferred to the probe optical signal **105**. However, when the wavelengths of these optical signals **105** and **110** are at a peak of the absorption spectrum, the modulation is transferred from the pump optical signal **110** to the probe optical signal **105**.
(17) After exiting the vapor cell **155**, the probe optical signal is reflected by the TFP **150** to an MTS PD **170**. That is, the TFP **150** reflects substantially all of the probe optical signal **105** to the MTS PD **170** so that little to none of the probe optical signal **105** passes towards the beam splitter **145**. In one embodiment, the propagation direction and polarization of the pump optical signal **110** are such that no pump light is directed towards the MTS PD **170**. Thus, the MTS PD **170** detects the probe optical signal **105** while the RAM PD **165** detects the pump optical signal **110**.
(18) In another embodiment, rather than directly receiving the probe optical signal **105**, the MTS PD **170** detects the fluorescence of the vapor cell **155** as the probe optical signal **105** pass through. When detecting the fluorescence of the vapor cell **155** using the PD **170**, the TFP **150** may be replaced with a simple polarizer since PD **170** can be located at various places along the cell and does not need to be in the beam path **105**.
(19) In this embodiment, a neutral density (ND) filter **135** (e.g., an optical attenuator) is placed in the probe optical signal **105** prior to the vapor cell **155**. The ND filter provides attenuation of the pump optical signal **110** after passing through the vapor cell **155**. This reduces the potential for stray reflections of the pump optical signal **110** off of the polarizer **130**, probe signal **105** input optics, or other components in the system from reaching the MTS PD **170**. Such reflections can

cause etalons and systematic errors in the signal measured by the MTS PD **170**. While ND filter **135** reduces the optical power of the probe optical signal **105** inside the vapor cell **155** as well, there is no disadvantage since the required power of the probe optical signal **105** is typically smaller than that of the pump optical signal **110**. Moreover stray reflections of the pump optical signal **110** that arise after passing through the vapor cell **155** are reduced twice by the ND filter **135** while the probe optical signal **105** is only reduced once.

(20) In one embodiment, the polarizer **130** can be a TFP or other polarization sensitive element which serves to direct the transmitted pump optical signal **110** away from the probe input optics (for example towards a beam dump, absorbing materials, or auxiliary photodiode) in order to avoid undesired pump light scatter from the polarizer **130** surface from reaching the MTS PD **170**.

(21) The electrical signals generated by the PDs **165** and **170** are provided to the control system **175** which uses these signals to generate a laser locking signal **180**, a first power adjustment signal **185**, an amplitude modulation correction signal **190**, and a second power adjustment signal **195**. The control system **175** can include hardware circuits (e.g., application specific integrated circuit(s), field programmable gate array(s), system on a chip, computing systems, controllers, etc.) software, firmware, or combinations thereof.

(22) In one embodiment, the laser locking signal **180** is used to lock a laser frequency to a peak of the absorption spectrum of the vapor in the vapor cell **155**. In one embodiment, the laser locking signal **180** is a laser adjustment signal that adjusts the wavelength of the optical source (or sources) that generates the pump optical signal **110** and the probe optical signal **105**. The control system **175** can sweep the carrier frequency or wavelength of lasers until they match a peak in the absorption spectrum of the vapor in the cell **155** or the fluorescence of the vapor cell **155**. To do so, the control system **175** can monitor the electrical signal generated by the MTS PD **170** which indicates the amount of modulation on the pump optical signal **110** that has transferred to the probe optical signal **105**.

(23) The control system **175** can continue to monitor the output of the MTS PD **170** to keep the carrier frequency or wavelength of the modulated pump optical signal **110** locked to the frequency corresponding to the peak in the absorption spectrum or fluorescence spectrum using the laser locking signal **180**. Thus, as environmental conditions change (e.g., change in temperature, humidity, etc.) or as the optical source ages, the spectroscopy system **100** can keep the optical source (not shown) outputting a signal at the desired wavelength or frequency which matches a peak in the absorption spectrum of the vapor cell **155**.

(24) However, the embodiments herein are not limited to using the laser locking signal **180** to adjust the wavelength of the optical source (e.g., laser) generating the pump optical signal **110**. The laser locking signal **180** can be used for a variety of applications including, e.g., high precision clocks, wavelength or frequency standards or gas analyzers.

(25) In addition to using a modulation portion of the output of the MTS PD **170** to generate the laser locking signal **180**, the control system **175** can also use the DC level the PD **170** outputs to generate the power adjustment signal **185** that adjusts the optical power of the probe signal **105**. For example, the control system **175** may include a direct current (DC) power servo that adjusts the DC power of the probe optical signal **105** using measurements obtained from the MTS PD **170**. In this manner, the control system **175** can use the output of the MTS PD **170** to perform two tasks: generate the laser locking signal **180** for performing MTS and generate the power adjustment signal **185** for the probe optical signal **105**. In one embodiment, the power adjustment signal **185** may be provided to an acousto-optic modulators (AOM), variable optical attenuator (VOA), or semiconductor optical amplifier (SOA) which controls the power of the probe optical signal **105**.

(26) The control system **175** can use the output of the RAM PD **165** to generate the amplitude correction signal **190** and the second power adjustment signal **195**. An undesirable amplitude modulation (e.g., RAM) may arise, for example, when modulating the frequency of the pump optical signal **110**. The control system **175** can use the modulation portion of the output of the PD

165 to measure the RAM in the pump optical signal **110** and then generate the amplitude correction signal **190** to mitigate the RAM. For example, the amplitude correction signal **190** may be provided to an AOM (not shown) upstream of the input pump optical signal **110** which then mitigates the RAM.

(27) In addition to using a modulation portion of the output of the RAM PD **165** to generate the amplitude correction signal **190**, the control system **175** can also use the DC level of the PD **165**'s output to generate the second power adjustment signal **195** that adjusts the optical power of the pump signal **110**. For example, the control system **175** may include a DC power servo that adjusts the DC power of the pump optical signal **110** using measurements obtained from the RAM PD **165**. In this manner, the control system **175** can use the output of the RAM PD **165** to perform two tasks: generate the amplitude correction signal **190** for mitigating RAM and generate the second power adjustment signal **195** for the pump optical signal **110**. In one embodiment, the second power adjustment signal **195** may be provided to an AOM, VOA, or SOA which controls the power of the pump optical signal **110**. In situations where the probe and pump optical signals **105**, **110** are generated using the same laser, the second power adjustment signal **195** can be used to adjust the power of both the probe and pump optical signals **105**, **110** by adjusting a SOA in the laser. A separate VOA or AOM can then be controlled using the first power adjustment signal **185** to adjust the power of the probe optical signal **105** so that a desired power ratio between the two optical signals is achieved.

(28) FIG. **2** is a flowchart of a method **200** for performing spectroscopy, according to one embodiment described herein. At block **205**, a control system (e.g., the control system **175** in FIG. **1**) measures a first optical signal using a first PD. For example, the first optical signal can be the pump optical signal **110** in FIG. **1** which is measured using the RAM PD **165**.

(29) At block **210**, the control system performs amplitude modulation correction and power correction using the measurement obtained from the first PD. For example, the measurements derived from the first PD can be used to generate the amplitude correction signal **190** and the power adjustment signal **195** in FIG. **1**. In one embodiment, the amplitude correction signal is used by an AOM to correct for RAM in the first optical signal, while the power adjustment signal controls the DC power of an optical source that generates the first optical signal—e.g., a SOA.

(30) At block **215**, the control system measures a second optical signal using a second PD. For example, the second optical signal may be the probe optical signal **105** in FIG. **1** which is detected by the MTS PD **170**.

(31) At block **220**, the control system performs laser locking and power correction using the measurements obtained from the second PD. In one embodiment, laser locking is part of an MTS-algorithm where the control system measures whether the modulation on the first optical signal (e.g., the pump optical signal) was transferred to the second optical signal (e.g., the probe optical signal) in a vapor cell. However, the embodiments herein are not limited to MTS and can be used in other types of spectroscopy.

(32) In one embodiment, the control system uses the measurements derived from the second PD to generate a laser locking signal and a power adjustment signal. The laser locking signal can adjust the wavelength of the first and second optical signals so the control system can detect a peak of the absorption spectrum of the vapor in the vapor cell through which the optical signals pass. For example, the control system can detect when the modulation on the first optical signal is transferred to the unmodulated second optical signal as the signals pass through the vapor cell. When the wavelengths of the optical signals are not at a peak of the absorption spectrum of the vapor in the vapor cell, little to none of the modulation in the first optical signal is transferred to the second optical signal. However, when the wavelengths of these optical signals are at a peak of the absorption spectrum, the modulation is transferred from the first optical signal to the second optical signal.

(33) The spectroscopy system can use the power adjustment signal derived from the second PD to

control the power of the second optical signal. In one embodiment, the control system compares the measurements from the first and second PDs to determine whether the powers of the first and second optical signals are at the desired values, and if not, use the power adjustment signal to adjust the power of the second optical signal until the desired value is obtained.
(34) In this manner, the method **200** describes techniques for using two PDs to perform four tasks: correcting modulation, adjusting the power of a first optical signal, performing laser locking, and adjusting the power of a second optical signal.
(35) FIG. **3** illustrates optical components in an optical bench for redirecting optical signals to PDs, according to one embodiment described herein. Specifically, FIG. **3** illustrates the beam splitter **145** and the TFP **150** which are tasked with reflecting the pump optical signal **110** and the probe optical signal **105** to the PDs **165** and **170**.
(36) As shown, the beam splitter **145** has a wedge shape where a surface **305** is not parallel with an anti-reflective (AR) coating **310**. Similarly, the TFP **150** also has a wedge shape where a surface **315** is not parallel with an AR coating **320**. Forming the beam splitter **145** and the TFP **150** in a wedge as shown in FIG. **3** reduces etalons that degrade performance. Thus, it may be advantageous to have wedge shaped optical components.
(37) In one embodiment, the surface **305** of the beam splitter **145** includes a non-polarizing beam splitter layer that is polarization insensitive. Further, arranging the non-polarizing beam splitter layer on the beam splitter **145**, which is before the TFP **150** (relative to the optical path of the pump optical signal), reduces the number of optics the probe optical beam passes through prior to reaching the MTS PD **170**. This may be advantages to reduce etalons and stray reflections.
(38) In one embodiment, the surface **315** of the TFP **150** includes a TFP layer that is disposed on the body of the TFP **150** opposite the side containing the AR coating **320**. The TFP layer can separate the probe optical signal (which has already passed through the vapor cell) from the pump optical signal, and deflect the probe optical signal to the MTS PD **170** shown in FIG. **1**.
(39) The bottom half of FIG. **3** illustrates arranging the beam splitter **145** and the TFP **150** at different angles in order to compensate for the wedge shape of each of these optical components. While the wedge shape is useful for reducing the undesirable etalons, it also deflects the optical signals in a non-parallel manner. That is, because of the wedge shape of the beam splitter **145**, a path **330** of the pump optical signal is not parallel with the path **335** of the pump optical signal after being transmitted through the beam splitter **145**. However, this deflection between the paths **330** and **335** can be compensated for or removed by appropriate selection of wedge angles and the optics angles (i.e., Θ.sub.1 and Θ.sub.2). As a result, the path **340** of the pump optical signal when exiting the TFP **150** is parallel with the path **330**.
(40) FIG. **3**, however, illustrates just one possible arrangement of the beam splitter **145** and the TFP **150** to compensate for the deflection caused by using wedge shaped optical components. In general, the beam splitter **145** and the TFP **150** can have any arrangement where the deflection caused by the pump optical signal passing through the wedge shape of the beam splitter **145** is compensated for by the TFP **150** such that a direction of propagation of the pump optical signal when entering the beam splitter **145** is parallel with a direction of propagation of the pump optical signal when exiting the TFP **150**.
(41) FIGS. **4**A-**4**C illustrate optical components in an optical bench for redirecting optical signals to PDs, according to one embodiment described herein. Specifically, FIGS. **4**A-**4**C illustrates using a unitary or integrated optical component to reflect the pump and probe optical signals to PDs, rather than using separate optical components as shown in FIGS. **1** and **3**. Put differently, the beam splitter **145** and the TFP **150** in FIG. **1** can be replaced by one of the optical components illustrated in FIGS. **4**A-**4**C.
(42) FIG. **4**A illustrates a unitary optical component that includes a beam splitter surface **405** and a TFP surface **430** (e.g., one example of a polarizer surface) disposed on opposite sides of a single optic. The pump optical signal enters at the beam splitter surface **405** (which can be uncoated or a

non-polarizing beam splitter coating). Some of the power of the pump optical signal is reflected by the beam splitter surface **405** and is detected by the RAM PD **165** as discussed above. The remaining portion of the pump optical signal passes through the component **400** and then exits at the TFP surface **430**. However, a portion of the pump optical signal is reflected at the TFP surface **430** and is labeled as the pump waste signal **420** (e.g., a waste beam) and exits the optical component **400** at the top side which has an AR coating **415**.
(43) In one embodiment, the width (in the X direction) of the optical component **400** is sufficiently large enough so that the pump waste signal **420** can exit at an AR coated surface **415**. Otherwise, the pump waste signal **420** may continue to reflect in the optical component **400** and cause diffuse scatter or glow which can be detected by the PD **165** or PD **170**, which is generally undesirable.
(44) After passing through the vapor cell, the probe optical signal **105** reaches the TFP surface **430** which reflects most of the signal to the MTS PD **170**. However, a portion of the probe optical signal **105** can pass through the surface **430**, reach the beam splitter surface **405** and is reflected downwards before exiting the component **400** at the bottom side with an AR coating **410**. This is labeled as the probe waste signal **425**. Like with the pump waste signal **420**, the width or thickness of the optical component **400** can be set so that these waste signals can exit at sides with the AR coatings **410** and **415** to prevent these signals from continuing to reflect in the optical component **400**, which can cause the component **400** to glow and be detected by the PDs **165** and **170**.
(45) FIG. **4**B illustrates a unitary optical component **450** that includes a beam splitter surface **455** and TFP surface **460** disposed on opposite sides. The pump optical signal enters at the beam splitter surface **455** (which may be uncoated or non-polarizing beam splitter coating). Some of the power of the pump optical signal is reflected by the beam splitter surface **455** and is detected by the RAM PD **165** as discussed above. The remaining portion of the pump optical signal passes through the component **450** and then exits at the TFP surface **460**. However, a portion of the pump optical signal is reflected at the TFP surface **460** and is labeled as the pump waste signal **475** and exits the optical component **450** at the bottom side which has an AR coating **470**.
(46) After passing through the vapor cell, the probe optical signal **105** reaches the TFP surface **460** which reflects most of the signal to the MTS PD **170**. However, a portion of the probe optical signal **105** can pass through the surface **460**, reach the beam splitter surface **455**, and be reflected downwards before exiting the component **450** at the bottom side with the AR coating **470**. This is labeled as the probe waste signal **465**. Like with the component **400** in FIG. **4**A, the width or thickness of the optical component **450** in FIG. **4**B can be set so that these waste signals (e.g., waste beams) exit at a side with an AR coating **470** to prevent these signals from continuing to reflect in the optical component **450**, which can cause the component **450** to glow and be detected by the PDs **165** and **170**.
(47) FIG. **4**C illustrates a unitary optical component **480** that includes an uncoated surface **485** and TFP surface **460** disposed on opposite sides. The body of the component **480** has the shape of a dove prism, but other shapes are possible.
(48) The pump optical signal **110** enters at the uncoated surface **485**. Some of the power of the pump optical signal **110** is reflected by the uncoated surface **485** and is detected by the RAM PD **165** as discussed above. The remaining portion of the pump optical signal **110** passes through the body of the component **480**, reflects off a bottom surface **490**, and then exits at the TFP surface **460** where it is directed to the vapor cell **155**. In one embodiment, there is total internal reflection when the pump optical signal **110** reflects off the bottom surface **490**. In one embodiment the uncoated surface **485** is replaced by a non-polarizing beam splitter coating.
(49) After passing through the vapor cell **155**, the probe optical signal **105** reaches the TFP surface **460** which reflects the signal to the MTS PD **170**, similar to what was discussed above.
(50) The bodies of the components **400**, **450**, and **480**, which define the shapes of these components, can include a transparent material such as glass. In these examples, the body of the optical component **400** forms a parallelogram or parallelepiped while the body of the optical

components **450** and **480** form a triangle or prism.
(51) Moreover, a portion or the entire bottom surface **490** of the optical component **480** can have an AR coating to permit waste signals to pass therethrough. That is, waste signals may pass from the uncoated surface **485** and the TFP surface **460** and exit the optical component **480** via the bottom surface **490**.
(52) FIG. **5**A illustrates integrating a vapor cell with an optical component for redirecting optical signals to PDs, according to one embodiment described herein. In one embodiment, the optical system **500** in FIG. **5**A can be used to replace the beam splitter **145**, TFP **150**, and vapor cell **155** in FIG. **1**. Stated differently, the optical components in FIG. **5**A can perform the same function as the beam splitter **145**, TFP **150**, and vapor cell **155** in FIG. **1**.
(53) The probe optical signal **105** and pump optical signal **110** are shown at the right of the figure. For example, the probe optical signal **105** may have already passed through the polarizer **130** and the ND filter **135** shown in FIG. **1**. The pump optical signal **110** may have already passed through the polarizer **125** in FIG. **1**. In addition, the pump optical signal **110** passes through a polarizer **505** which alters the polarization of the pump optical signal **110**. As a result, a TFP surface **510** reflects a portion of the pump optical signal **110** down towards the RAM PD **165** while the remaining portion of the signal **110** enters an optical window **515** of the vapor cell **550**.
(54) As shown, the vapor cell **550** includes windows **515** and **520**, which have a wedge shape, on opposite ends where the probe and pump optical signals **105**, **110** enter and exit the vapor cell **550**. Although not shown in FIG. **1**, the vapor cell **155** may also include optical windows at its ends. However, in FIG. **5**A, the window **515** is modified to perform the functions of the beam splitter **145** and the TFP **150** in FIG. **1**. For example, the window **515** may be thicker (or wider) than the window on the vapor cell **155** in FIG. **1**. Further, the window **515** includes the TFP surface **510** for reflecting a portion of the pump optical signal **110** to the RAM PD **165** and reflecting the probe optical signal **105** (after passing through the vapor cell **550**) from the TFP surface **510**. Further, the window **515** includes an AR coating **530** on the top side to permit the reflected probe optical signal **105** to pass through the window **515** and reach the PD **170**. This AR coating **530** also provides an interface for the probe optical signal **105** to enter the window **515**.
(55) Thus, FIG. **5**A illustrates an optical system **500** where a window **515** of the vapor cell **550** can be modified to replace the beam splitter **145** and the TFP **150** in FIG. **1** (or the unitary optical components **400**, **450**, and **480** in FIGS. **4**A-**4**C).
(56) In one embodiment, the AR coatings **530** are applied to the windows **515** and **520** before they are attached to the vapor cell **550**. In one embodiment, the coating or film used to form the TFP surface **510** can be added after the cell **550** is fabricated to avoid exposure to a high temperature process while sealing the vapor cell **550**.
(57) While FIG. **5**A illustrates mirrors **160** disposed to the left of the window **520**, in other embodiments, a high reflection (HR) mirror can be added to the left side of the window **520** for reflecting the optical signals. For example, the probe and pump optical signals **105**, **110** can be directed into the vapor cell **550** at an angle and then reflect off the HR mirror of the window **520**. Doing so removes extra components (e.g., the mirrors **160**) and simplifies the overall system.
(58) FIG. **5**B illustrates another embodiment where the mirrors **160** in FIG. **5**A can be omitted. In optical system **560**, the second cell window **520** and mirrors **160** depicted in FIG. **5**A can be replaced with a single reflector **565**. The reflector **565** can perform the same function as the window **520** and mirrors **160** serving as both the vapor cell seal and beam steering to enable the probe and pump optical signals **105**, **110** to pass multiple times through the vapor cell.
(59) In one embodiment, the reflector **565** is a glass prism configured for total internal reflection, such as a roof prism, allowing a similar function to the mirrors **160** in FIG. **5**A. In this case, the beam alignment occurs in such a manner that the probe optical signal **105** and pump optical signal **110** pass through fewer optical surfaces in the system which is advantageous to reduce etalons, diffuse scatter, and system complexity. In another embodiment the surfaces of the reflector **565**

have a thin film or dielectric coating to perform the mirror function.
(60) In one embodiment the reflector **565** reduces undesired reflections and etalons in the optical system **560** by including an AR coated surface **530** on the reflector input side facing the vapor cell **550** and by tilting the reflector **565** to avoid normal incidence.
(61) FIG. **6** illustrates a spectroscopy system **600** with an optical bench **615**, according to one embodiment described herein. In this example, the optical bench **615** includes many of the same components as the optical bench **115** in FIG. **1**, which are indicated by using the same reference numbers. However, unlike in FIG. **1**, the optical bench **615** includes a beam splitter **645** and TFP **650** which both the probe optical signal **105** and pump optical signal **110** pass through, along with a waveplate **640**. Passing both optical signals through the same beam splitter and TFP can allow for more compact geometries and reduce complexity.
(62) The beam splitter **645** crosses the optical path of both the probe and pump optical signals **105**, **110**. Doing so results in the beam splitter **645** reflecting a portion of the probe optical signal **105** to an optional probe input PD **605**. Though not required, the output of the probe input PD **605** can be provided to the control system (not shown) to monitor and adjust the power of the probe optical signal **105**, monitor the probe power control performance in an out-of-loop manner, or suppress residual intensity noise (RIN) on the probe optical signal.
(63) Due to miniaturization, or because the spacing between the pump and probe optical signals has been reduced, FIG. **6** illustrates that the TFP **650** can also be in the path of the probe optical signal **105** before this signal reaches the vapor cell **155**. Moreover, the TFP **650** is also in the path of the probe optical signal **105** after it passes through the vapor cell **155**. As such, the TFP **650** can perform the function discussed above where it reflects the probe optical signal **105** to the MTS PD **170**.
(64) In sum, the optical bench **615** allows additional measurements to be captured (e.g., the measurement obtained using the probe input PD **605**) and a smaller optical bench **615**, without adding any additional components to the optical bench **615**. Although not shown, FIG. **6** can also include a ND filter which is disposed between, e.g., the TFP **650** and the waveplate **640**.
(65) FIG. **7** illustrates a spectroscopy system with an optical bench **115**, according to one embodiment described herein. The optical bench **115** in FIG. **7** is the same as the optical bench **115** in FIG. **1**, and can have any of the alternative arrangements discussed above. In addition to having the optical bench **115**, the system **700** includes a splitter **705** that splits the power in the probe optical signal **105** so that a portion of this signal **105** is transmitted to a reference PD **710**. The output of the reference PD **710** is coupled to the control system **175** and can be used to reduce noise in the probe optical signal **105**.
(66) In one embodiment, the optical source used to generate the probe optical signal **105** can have significant residual intensity noise (RIN) in a frequency band of interest. To suppress the impact of RIN, a reference PD **710** can be used in combination with the control system **175** and MTS PD **170**. This system could also be useful in the optical systems illustrated in FIGS. **1** and **6**.
(67) Moreover, the embodiments described above can be applied to spectroscopy systems that use an unmodulated reference optical signal to mitigate noise. In those embodiments, the unmodulated reference optical signal is transmitted through the same vapor cell as the modulated pump optical signal and the probe optical signal as discussed above. Examples of this are described in NOISE MITIGATION IN VAPOR CELL SPECTROSCOPY, U.S. patent application Ser. No. 17/933,735, which is hereby incorporated by reference.
(68) The preceding description is provided to enable any person skilled in the art to practice the various embodiments described herein. The examples discussed herein are not limiting of the scope, applicability, or embodiments set forth in the claims. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments. For example, changes may be made in the function and arrangement of elements discussed without departing from the scope of the disclosure. Various

examples may omit, substitute, or add various procedures or components as appropriate. For instance, the methods described may be performed in an order different from that described, and various steps may be added, omitted, or combined. Also, features described with respect to some examples may be combined in some other examples. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method that is practiced using other structure, functionality, or structure and functionality in addition to, or other than, the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

(69) As used herein, the word "exemplary" means "serving as an example, instance, or illustration." Any aspect described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other aspects.

(70) As used herein, a phrase referring to "at least one of" a list of items refers to any combination of those items, including single members. As an example, "at least one of: a, b, or c" is intended to cover a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c).

(71) As used herein, the term "determining" encompasses a wide variety of actions. For example, "determining" may include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, "determining" may include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, "determining" may include resolving, selecting, choosing, establishing and the like.

(72) The methods disclosed herein comprise one or more steps or actions for achieving the methods. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is specified, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims. Further, the various operations of methods described above may be performed by any suitable means capable of performing the corresponding functions. The means may include various hardware and/or software component(s) and/or module(s), including, but not limited to a circuit, an application specific integrated circuit (ASIC), or processor. Generally, where there are operations illustrated in figures, those operations may have corresponding counterpart means-plus-function components with similar numbering.

(73) As will be appreciated by one skilled in the art, the embodiments disclosed herein may be embodied as a system, method, or computer program product. Accordingly, embodiments may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a "circuit," "module" or "system." Furthermore, embodiments may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

(74) Program code embodied on a computer readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

(75) Computer program code for carrying out operations for embodiments of the present disclosure may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the "C" programming language or similar programming languages. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote

computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

(76) Aspects of the present disclosure are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatuses (systems), and computer program products according to embodiments presented in this disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the block(s) of the flowchart illustrations and/or block diagrams.

(77) These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other device to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the block(s) of the flowchart illustrations and/or block diagrams.

(78) The computer program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process such that the instructions which execute on the computer, other programmable data processing apparatus, or other device provide processes for implementing the functions/acts specified in the block(s) of the flowchart illustrations and/or block diagrams.

(79) The flowchart illustrations and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments. In this regard, each block in the flowchart illustrations or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustrations, and combinations of blocks in the block diagrams and/or flowchart illustrations, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

(80) The following claims are not intended to be limited to the embodiments shown herein, but are to be accorded the full scope consistent with the language of the claims. Within a claim, reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f) unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.” All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims.

Claims

1. A spectroscopy system, comprising: a vapor cell configured to receive a probe optical signal and a pump optical signal, wherein optical paths of the probe and pump optical signal in the vapor cell at least partially overlap, wherein the probe optical signal is an unmodulated optical signal before entering the vapor cell, wherein the pump optical signal is a modulated optical signal; a first photodiode configured to receive a portion of the pump optical signal before passing through the vapor cell; a second photodiode configured to at least one of (i) receive a portion of the probe optical signal after passing through the vapor cell or (ii) detect a fluorescence of the vapor cell; and a control system configured to: perform amplitude correction and power adjustment on the pump optical signal using measurements obtained from the first photodiode, and perform power adjustment on the probe optical signal and laser locking using measurements obtained from the second photodiode.
2. The spectroscopy system of claim 1, further comprising: a beam splitter surface configured to reflect a portion of the pump optical signal to the first photodiode and transmit the remaining portion of the pump optical signal to the vapor cell; and a polarizer surface configured to receive the probe optical signal after passing through the vapor cell and reflect the probe optical signal to the second photodiode, wherein the remaining portion of the pump optical signal passes through the polarizer surface before reaching the vapor cell.
3. The spectroscopy system of claim 2, wherein the beam splitter surface is disposed on a first wedge shaped optical component and the polarizer surface is a thin film polarizer (TFP) disposed on a second wedge shaped optical component.
4. The spectroscopy system of claim 3, wherein a deflection on the pump optical signal caused by passing through the wedge shape of the first wedge shaped optical component is compensated for by the second wedge shaped optical component such that a direction of propagation of the pump optical signal when entering the first wedge shaped optical component is parallel with a direction of propagation of the pump optical signal when exiting the second wedge shaped optical component.
5. The spectroscopy system of claim 2, wherein the beam splitter surface and the polarizer surface are disposed on opposite sides of a unitary optical component having a shape of a parallelepiped, wherein the unitary optical component comprises respective anti-reflective (AR) coatings through which waste beams corresponding to the probe and pump optical signals pass when exiting the unitary optical component.
6. The spectroscopy system of claim 2, wherein the beam splitter surface and the polarizer surface are disposed on opposite sides of a unitary optical component having a shape of a prism, wherein the unitary optical component comprises an AR coating through which waste beams corresponding to the probe and pump optical signals pass when exiting the unitary optical component.
7. The spectroscopy system of claim 2, wherein the beam splitter surface and the polarizer surface are disposed on opposite sides of a unitary optical component having a shape of a dove prism.
8. The spectroscopy system of claim 2, wherein the beam splitter surface is disposed on a first optical component and the polarizer surface is a thin film polarizer (TFP) disposed on a second optical component, wherein at least a portion of the probe optical signal passes through both the first and second optical components before reaching the vapor cell, wherein at least a portion of the pump optical signal passes through both the first and second optical components before reaching the vapor cell.
9. The spectroscopy system of claim 1, further comprising: a window contacting the vapor cell and arranged to receive the pump and probe optical signals, wherein the window comprises a polarizer surface that reflects the portion of the pump optical signal to the first photodiode and reflects the portion of the probe optical signal to the second photodiode.

10. The spectroscopy system of claim 9, wherein the window comprises an AR coating through which one of the reflected portions of the pump or probe optical signal passes before reaching the first or second photodiode.
11. The spectroscopy system of claim 10, further comprising: a reflector disposed on a side of the vapor cell opposite the window, the reflector configured to seal the vapor cell and enable the pump and probe optical signals to pass multiple times through the vapor cell.
12. The spectroscopy system of claim 1, further comprising: an optical attenuator configured to attenuate the pump optical signal after passing through the vapor cell and attenuate the probe optical signal before passing through the vapor cell.
13. A spectroscopy system, comprising: a vapor cell configured to receive a probe optical signal and a pump optical signal, wherein optical paths of the probe and pump optical signal in the vapor cell at least partially overlap; a first photodiode configured to receive a portion of the pump optical signal before passing through the vapor cell; a second photodiode configured to at least one of (i) receive a first portion of the probe optical signal after passing through the vapor cell or (ii) detect a fluorescence of the vapor cell; a third photodiode configured to receive a second portion of the probe optical signal before passing through the vapor cell; and a control system configured to: perform amplitude correction and power adjustment on the pump optical signal using measurements obtained from the first photodiode; perform laser locking using measurements obtained from the second photodiode; and reduce noise on the probe optical signal using measurements obtained from the third photodiode.
14. The spectroscopy system of claim 13, further comprising: a beam splitter surface configured to reflect a portion of the pump optical signal to the first photodiode and transmit the remaining portion of the pump optical signal to the vapor cell; and a polarizer surface configured to receive the probe optical signal after passing through the vapor cell and reflect the probe optical signal to the second photodiode, wherein the remaining portion of the pump optical signal passes through the polarizer surface before reaching the vapor cell.
15. The spectroscopy system of claim 14, wherein the beam splitter surface is disposed on a first wedge shaped optical component and the polarizer surface is a thin film polarizer (TFP) disposed on a second wedge shaped optical component.
16. The spectroscopy system of claim 15, wherein a deflection on the pump optical signal caused by passing through the wedge shape of the first wedge shaped optical component is compensated for by the second wedge shaped optical component such that a direction of propagation of the pump optical signal when entering the first wedge shaped optical component is parallel with a direction of propagation of the pump optical signal when exiting the second wedge shaped optical component.
17. The spectroscopy system of claim 14, wherein the beam splitter surface and the polarizer surface are disposed on opposite sides of a unitary optical component having a shape of a parallelepiped, wherein the unitary optical component comprises respective AR coatings through which waste beams corresponding to the probe and pump optical signals pass when exiting the unitary optical component.
18. The spectroscopy system of claim 14, wherein the beam splitter surface and the polarizer surface are disposed on opposite sides of a unitary optical component having a shape of a prism, wherein the unitary optical component comprises an AR coating through which waste beams corresponding to the probe and pump optical signals pass when exiting the unitary optical component.
19. The spectroscopy system of claim 14, wherein the beam splitter surface and the polarizer surface are disposed on opposite sides of a unitary optical component having a shape of a dove prism.
20. The spectroscopy system of claim 14, wherein the beam splitter surface is disposed on a first optical component and the polarizer surface is a TFP disposed on a second optical component,

wherein at least a portion of the probe optical signal passes through both the first and second optical components before reaching the vapor cell, wherein at least a portion of the pump optical signal passes through both the first and second optical components before reaching the vapor cell.

21. The spectroscopy system of claim 13, further comprising: a window contacting the vapor cell and arranged to receive the pump and probe optical signals, wherein the window comprises a polarizer surface that reflects the portion of the pump optical signal to the first photodiode and reflects the first portion of the probe optical signal to the second photodiode.

22. The spectroscopy system of claim 21, wherein the window comprises an AR coating through which one of the reflected portions of the pump or probe optical signal passes before reaching the first or second photodiode.

23. A method, comprising: measuring a pump optical signal using a first photodiode before the pump optical signal has passed through a vapor cell, wherein the pump optical signal is a modulated optical signal; performing amplitude modulation correction and power adjustment on the pump optical signal using measurements obtained from the first photodiode; at least one of (i) measuring a probe optical signal or (ii) detecting a fluorescence of the vapor cell using a second photodiode after the probe optical signal has passed through the vapor cell, wherein the probe optical signal is an unmodulated optical signal before entering the vapor cell; and performing power adjustment on the probe optical signal and laser locking using measurements obtained from the second photodiode.